Normalized C-zero output curve membership

Derived the linear quotient and output curve equation from actual polynomial premises. Assume a monic polynomial f of degree five with coefficient four zero, two monic quadratic supports u1,u2, valid input residual divisibilities, B nonzero and C zero. The independently reviewed interpolation result supplies l=v1-(A/B)u1=v2-(A/B)u2 and u1*u2 divides f-l^2. New results derive f-l^2=u1*u2*(X-x), x=q1+q2+(A/B)^2, and y^2=f(x) for y=-l(x). No quotient formula is assumed as a premise.

The proof bounds l at degree two, hence f-l^2 has degree five. Each support has degree two, so the quotient has degree one. Its leading coefficient is one by the actual coefficient-five equation; coefficient four forces its constant term to -(q1+q2+(A/B)^2). Evaluation at x supplies the output equation. These are normalized Field-algebra results; smoothness is not needed for this component.

Twelve main declarations and ten fixed control/witness declarations pass, with 22 separate axiom readbacks using only propext, Classical.choice and Quot.sound. Final output2, quotient5, point2, controls3 and readback1 have native exit zero and unchanged input bindings. Controls instantiate the full original-input theorem with rational u1=X^2+X+2, u2=X^2+1, v1=4X+5, v2=X+2. Explicit input witnesses and monic quintic f=X^5-13X^3-22X^2-10X-19 give x=10 and y=291. Smoothness of this fixture is not asserted. A nonzero-fourth-coefficient control records the shape boundary.

Source links: projective_law_gate.rs1002-1023 gives the C-zero weighted payload. At normalized unit scales its ordinate is -l(x). projective_state_collector.rs137-142 derives shift=-old[4]/(5*lead) and asserts normalized[5]=1 and normalized[4]=0; projective_law_gate.rs466-471 has the same normalization checks. This proof assumes those polynomial shape properties; it does not prove that transform construction. Weighted coordinate-scale transport, nonzero weights, machine word correspondence, compiled execution, divisor-class addition and whole-collector induction remain separate obligations.

Failures are retained: probe1 requested unavailable lemma names; output1 passed an implicit theorem as a function; quotient1 used unavailable linear_combination and insufficient coefficient simplification; quotient2 reached the default heartbeat limit; quotient3 omitted explicit coefficients of X; quotient4 needed simplification of multiplication by one; point1 and controls1 required explicit evaluation unfolding. Superseded successful controls2 remains. Quotient final check uses a declared two-million-heartbeat limit within the unchanged 120-second process limit. No sampled scientific runs or repository edits occurred. This producer packet awaits independent review and retains its imported compiled dependency trust boundary.

Artifact references Output.lean; Quotient.lean; CurvePoint.lean; PointControls.lean; status.json. The editable diagram is flow.svg. Source assumptions and failures are recorded in REPORT.md.